

Serving

OpenAI-compatible
API gateway

Continuous-batching
decode engine

Session manager
(residency, lifecycle)

MLXDiffKVWrapper — generate(): chunked prefill + decode loop

MLXQwenModel — native KV cache (prefill only) + prefill→decode release

Patched Qwen2 attention

prefill ($L > 1$)

decode ($L = 1$)

Prefill path

block-sparse causal attention
over the native cache

capture K/V → stream-compress
blocks leaving the window

write

Decode path

ingest token into dense window
(+ flush eligible block)

route top-K → fused low-rank +
exact-residual + recency attention

read (no decompress)

MLXKVBlockManager — per-session KV store (×28 layers)
dense recency window $[H_{kv}, 1, 280, d]$ fp16 (newest 1,024 tokens)
+ compressed pool $\{U, V_K, V_V, \text{anchors}, \text{residuals}, \text{min/max}\}$ × 256 blocks